

BChemXtract 1.4.1 vs Open Babel 3.1.1

Structure-extraction performance over the BChemXtract `.cdx` test corpus

Generated 2026-09-09 · 225 ChemDraw binary files under `testset/corpus` · BChemXtract 1.4.1 (JDK 25) vs Open Babel 3.1.1 · identical corpus, both deduplicating per file on InChIKey

At a glance

CHEMICAL QUALITY 98% vs 14% Real InChIs: BChemXtract vs Open Babel	UNEXPANDED ALIASES 23 vs 617 Leftover <code>*</code> atoms in the unique set	SPEED 51 vs 55 ms Per file, excluding Open Babel's 44 s outlier
---	---	--

Comparison

METRIC	BCHEMXTRACT 1.4.1	OPEN BABEL 3.1.1
COVERAGE		
Files in corpus	225	225
Files yielding structures	203	224
Empty (0 found)	21	0
Failures	1	1
YIELD & CHEMICAL QUALITY		
Unique substances	1360	1317
Real InChIs generated	1336 (98.2%)	187 (14.2%)
SMILES fallback (no InChI)	24	1130
Structures with unexpanded alias (<code>*</code>)	23	617
Fully expanded structures	1337	700
Raw structures before dedup	1979 fragments	1864
SPEED		
Total time	11.5 s	56.7 s
Average per file	51 ms	252 ms
Average per file, outlier excluded	51 ms	55 ms
Throughput	19.6 files/s	4.0 files/s

The 4.9× total-time gap is a single file. Open Babel spends 44.3 s of its 56.7 s on `volumeTest/bug/m17121576-9.cdx` — the same file BChemXtract v1.0 hung on indefinitely. Excluding it, Open Babel runs 224 files in 12.4 s (55 ms/file), essentially level with BChemXtract's 51 ms/file. The earlier report's "12–17× slower, dominated by slow InChI generation" does not hold on this run; the defensible speed claim is *comparable per file, with one pathological outlier*.

Key findings

- **Chemistry — the decisive difference.** BChemXtract produces a genuine InChI for 1336 of 1360 unique substances (98.2%). Open Babel manages 187 of 1317 (14.2%) and leaves 617 structures carrying wildcard `*` atoms, because it cannot expand ChemDraw abbreviations and S-groups. Most of its output is wildcard-bearing SMILES rather than identifiable substances.
- **Speed is close, once the outlier is separated.** 51 ms vs 55 ms per file. The order-of-magnitude claim in the 2026-07-08 report is an artifact of one file plus a different machine — see below.
- **Absolute times are not comparable to the earlier report.** This machine runs BChemXtract about 1.5× slower (51 ms/file vs 34 ms) and Open Babel about 1.6× faster (252 ms/file vs 401 ms) than the 2026-07-08 box. Quality figures are hardware-independent; timing ratios across reports are not.
- **BChemXtract's alias count is no longer zero.** v1.1.3 reported 0 unexpanded aliases and 1 SMILES fallback; 1.4.1 reports 23 and 24. The corpus also grew from 200 to 225 files, so this may be entirely new fixtures rather than a regression — it has not been attributed and should be before the "0 leftover `*` atoms" headline is reused.
- **Shared hard failure.** Both tools fail on `volumeTest/vol20Test/m31717432-graphical-abstract.cdx` — it is not a valid binary CDX (no ChemDraw header). BChemXtract: *IOException: Header not recognized*; Open Babel: *Invalid file, no ChemDraw Header*.
- **Coverage difference is semantic.** The 21 files BChemXtract reports as empty are S-group, copolymer and Markush fixtures it deliberately does not resolve into discrete substances. Open Babel emits raw structures for them, inflating its file-coverage count.

Conclusion

On this 225-file ChemDraw corpus, BChemXtract 1.4.1 decisively outperforms Open Babel 3.1.1 on extraction **quality**: 98.2% real InChIs against 14.2%, and 23 unresolved wildcards against 617. On **speed** the two are comparable per file (51 ms vs 55 ms); Open Babel's 4.9× worse total is one pathological file absorbing 44 s. For identifiable, InChI-grade substance extraction from ChemDraw files BChemXtract remains the clear choice, but the case rests on chemical resolution, not throughput.

Methodology. BChemXtract path: `CDXReader.readDocument` → `SubstanceExtractor.extractUnique(doc, info, false)`; "real InChI" = non-empty `BCXSubstance.getInchi()`; "unexpanded alias" = SMILES containing `*`. Open Babel path: `obabel file.cdx -ocan`, dedup on canonical SMILES, then `obabel -ismi -oinchikey`; "unique" = distinct InChIKeys + structures with no InChI, deduped on SMILES. Scope is `.cdx` only, matching Open Babel's reach; BChemXtract over the full `.cdx` + `.cdxml` corpus (246 files) yields 1415 substances / 1387 real InChIs in 12.6 s. Reproduce with `testset/run-bchemxtract-performance.sh` and `testset/run-openbabel-performance.sh`. Build: JDK 25 (Temurin 25.0.4.1+1), single warm JVM per run; Open Babel 3.1.1 from Ubuntu noble/universe, one process pair per file.